

Orientation and exchange statistics in a Golden six-mode transfer

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Abstract

Exchange statistics determine which information a many-particle interferometer can retain about its one-particle transfer. We study this distinction in an exact six-mode transfer constructed from the two three-dimensional eigenspaces of a real symmetric conference matrix. For a diagonal path control D_x , the postselected block

$$K_T(x) = Q_{T,-}^\top D_x Q_{T,+}$$

has filled three-fermion amplitude $\det K_T(x)$, proportional to a signed Joubert coordinate on the Segre cubic. We show that this orientation has no intrinsic permanent counterpart: permanents depend on the chosen input and output port frames, whereas the basis-independent bosonic observable is the symmetric-cube trace $h_3(K_T^\dagger K_T)$, a function only of the singular values. At each of twenty balanced controls, all six transfers have squared singular spectrum $\{1/5, 4/5, 4/5\}$, giving $h_3 = 313/125$ and exterior-cube probability $e_3 = 16/125$; the other 44 Boolean controls have zero fermionic amplitude. Calibrated determinant signs form a distance-six simplex code supporting economical three-, five-, and ten-cut readout, and a rational lossy control realizes an exact chiral anomaly-free charge vector projectively. We give a six-mode optical compilation and quantitative certification thresholds. Present photonic components support coherent transfer tomography and bosonic controls; direct three-fermion emulation additionally requires a totally antisymmetric three-photon qutrit resource. The result is a theory and design-limit analysis, not a report of a built device or a dynamical gauge-theory simulation.

1 Introduction

Let K be a three-input, three-output block of a linear one-particle transfer. If all three inputs and outputs are occupied, the fermionic transition amplitude is $\det K$, while the collision-free bosonic amplitude is $\text{per } K$. This familiar distinction does not by itself say whether the two amplitudes encode the same geometric information. The answer depends on which changes of input and output coordinates belong to the physical data.

We consider a six-mode transfer obtained from a real symmetric conference matrix C of order six, so that

$$C^\top = C, \quad C_{ii} = 0, \quad C_{ij} \in \{\pm 1\}, \quad C^2 = 5I.$$

The two eigenspaces of C , with eigenvalues $\pm\sqrt{5}$, both have dimension three. A diagonal path control couples them through a three-by-three block K . Six coherently marked conference matrices produce six such blocks. Their determinants are the six classical Joubert coordinates and satisfy the two equations of the Segre cubic. This identification, including

its invariant-theoretic provenance, is developed in the companion operator paper; here we ask what it means under symmetric and antisymmetric exchange statistics.

The central point is an orientation boundary. The Golden data determine the two three-dimensional port spaces but not ordered orthonormal frames inside them. Changing those frames acts on K from the left and right by orthogonal matrices. The determinant retains an orientation sign. The permanent does not descend through this action. Consequently the intrinsic bosonic companion is not a second signed Segre coordinate: it is a spectral quantity on the symmetric cube.

Theorem 3.1 states this distinction together with the exact balanced benchmark. The remaining sections separate intrinsic observables from calibrated ones, give a short arithmetic application, and identify the boundary of a photonic implementation. General boson–fermion complementarity in linear interferometers is due to Jabbour and Cerf [1]; its extension to partially distinguishable particles is due to Robbio, Jabbour, and Cerf [2]. The present contribution is the exact Golden specialization, the port-gauge obstruction, and the associated measurement and resource package.

2 The Golden transfer

For one member C_T of the marked conference family, let $Q_{T,+}$ and $Q_{T,-}$ be matrices whose columns are ordered orthonormal frames for the $\pm\sqrt{5}$ -eigenspaces. For $x = (x_0, \dots, x_5)$, put

$$D_x = \text{diag}(x_0, \dots, x_5), \quad K_T(x) = Q_{T,-}^\top D_x Q_{T,+}.$$

When $|x_i| = 1$, the six-mode transfer

$$U_T(x) = [Q_{T,+} \ Q_{T,-}]^\top D_x [Q_{T,+} \ Q_{T,-}]$$

is orthogonal. Values $|x_i| < 1$ describe a diagonal filter followed by postselection.

The six determinant amplitudes satisfy

$$Z_T(x) = \varepsilon_T, 10\sqrt{5} \det K_T(x), \quad \varepsilon_T \in \{\pm 1\},$$

with signs fixed by the marked orientations. They obey

$$\sum_T Z_T(x) = 0, \quad \sum_T Z_T(x)^3 = 0.$$

These are the Segre equations in Joubert coordinates. The equations and the outer S_6 -action are classical invariant theory [3]; the point used below is their realization as six amplitudes of one marked transfer.

3 Why the determinant retains orientation

Write $H = K^\dagger K$. For the intrinsic comparison, define

$$B_{\text{sym}}(K) = \text{tr}(\text{Sym}^3 H) = h_3(H), \quad F_{\text{ext}}(K) = \text{tr}\left(\bigwedge^3 H\right) = e_3(H) = \det H.$$

The first expression is the sum of all-transferred probabilities, averaged over the choice of normalized three-boson input occupation only after division by the ten-dimensional symmetric-cube input space. Thus the operational uniform-input average is $B_{\text{sym}}/10$. The second expression is the filled three-fermion probability.

Theorem 3.1 (Exchange-statistics orientation boundary). *Let $K_T(x)$ be a Golden transfer block.*

(i) *A change of ordered orthonormal port frames sends*

$$K \mapsto R_-^\top K R_+, \quad R_\pm \in O(3).$$

The determinant changes by $\det(R_-)\det(R_+)$, while the permanent does not define a scalar on the resulting double orbit.

(ii) *Every scalar intrinsic under the $O(3) \times O(3)$ port action factors through the singular values of K . Under $SO(3) \times SO(3)$, the sign of $\det K$ is the additional orientation label.*

(iii) *At every balanced control with three entries $+1$ and three entries -1 , for each of the six marked protocols,*

$$\text{spec } H = \left\{ \frac{1}{5}, \frac{4}{5}, \frac{4}{5} \right\}.$$

Consequently

$$B_{\text{sym}} = \frac{313}{125}, \quad F_{\text{ext}} = \frac{16}{125}, \quad B_{\text{sym}} - F_{\text{ext}} = \frac{297}{125}.$$

All intrinsic bosonic scalars are therefore blind to the marked Segre node and its orientation on this balanced family.

Proof. Part (i) follows directly from changing the two orthonormal frames. The determinant transformation law is multiplicative. The permanent has no corresponding covariance: for $K = I_3$, rotating the first two columns by an angle θ changes its value to $\cos(2\theta)$, although the singular values and $|\det K|$ remain fixed.

The real singular-value decomposition classifies the $O(3) \times O(3)$ double orbits, proving (ii). Restricting to the two special orthogonal groups retains the determinant sign. For (iii), the Golden conference identity gives the displayed spectrum on every balanced cut. The complete symmetric functions then give

$$h_3\left(\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right) = \frac{313}{125}, \quad e_3\left(\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right) = \frac{16}{125}.$$

Finally, the degree-three symmetric-function identity $h_3 - e_3 = p_1 p_2$ gives $297/125$. $\square$

The theorem does not say that calibrated bosonic measurements lose all control information. Once physical port axes are fixed, a permanent is an ordinary measurable amplitude. It says that such a value belongs to the calibrated apparatus rather than to the unmarked Golden operator.

4 Balanced controls and calibrated readout

There are 64 Boolean controls $x \in \{\pm 1\}^6$. The twenty balanced controls are the common nonzero determinant optimum. For each one,

$$|\det K_T| = \frac{4}{5\sqrt{5}}, \quad P_F = |\det K_T|^2 = \frac{16}{125}.$$

The other 44 controls have zero filled-fermion amplitude. This gives a sharp Boolean boundary rather than a numerical search criterion.

Freeze the port frames by a reproducible pivot convention. In that gauge the collision-free bosonic probabilities on balanced controls take the values

$$P_B(111 | 111) \in \left\{ \frac{16}{3125}, \frac{36}{3125}, \frac{64}{3125} \right\}.$$

These values are controls, not invariants. Coherent access to the calibrated amplitudes distinguishes all twenty oriented masks; squaring pairs antipodal controls and loses the overall orientation.

Across the ten projective balanced cuts, the six determinant sign words have pairwise Hamming distance six. Three selected signs identify a marked protocol once Golden membership is already known. Five signs and magnitudes jointly test membership and identify the protocol. Measuring all ten signs supports soft matched-filter decoding; hard nearest-word decoding corrects two sign errors. No relative sign word fixes the global orientation without a coherent port reference.

The universal complementarity identities for determinants and permanents of Jabbour and Cerf [1] provide a useful calibrated consistency check. They do not distinguish the Golden family by themselves.

5 An exact arithmetic instance

The rational diagonal filter

$$x = \left(1, \frac{7}{13}, \frac{1}{7}, -\frac{1}{5}, -\frac{1}{2}, -1 \right)$$

gives

$$Z(x) = \frac{72}{455}(11, -10, -8, 5, 4, -2).$$

The integer vector on the right has vanishing sum and vanishing sum of cubes, so it is an anomaly-free chiral charge vector for six Weyl fermions. General parametrizations of Abelian anomaly equations are known [4, 5]; the contribution here is the operator filter and its exact postselection cost. In the present normalization,

$$P_{F,T} = \frac{(72/455)^2 q_T^2}{500}, \quad q = (11, -10, -8, 5, 4, -2).$$

This construction realizes arithmetic coordinates satisfying the anomaly equations. It contains no gauge field, regulator, anomalous current, or gauge-theory dynamics. The extension to two $U(1)$ factors is governed by lines on the Segre cubic and has its own established algebraic geometry [6]; it is outside the scope of this paper.

6 Photonic design limit

The balanced six-mode transfer needs at most fifteen tunable two-mode cells. A filtered implementation uses two such meshes and six attenuators. Coherent transfer characterization fixes the input and output phase gauges and infers the sign of the reconstructed real determinant. Six-mode coherent transfer tomography of this kind has been demonstrated [10], and programmable twelve-mode processors have operated with three photons [11].

This feasible precursor supports two distinct tests:

1. reconstruct the six-mode transfer coherently and test the determinant signs and the linear and cubic Segre residuals;

2. inject ordinary photons and measure the calibrated bosonic controls, optionally upgrading to the ten-input symmetric-cube average.

The inferred determinant sign in the first test is not a direct measurement of a many-fermion phase.

The photonic fermion emulator of Matthews et al. [7] sends an entangled resource through identical copies of a one-particle process. For three particles in this setting it requires a totally antisymmetric state of three photonic qutrits. Goyal et al. [8] give a linear-optical preparation proposal for this state. The design evaluated here treats a demonstrated, characterized source of that resource as an unmet input. A three-qutrit GHZ state, although experimentally accessible [9], has different exchange symmetry and does not substitute for it.

The determinant perturbation bound

$$|\det(K + E) - \det K| \leq 3\|E\|_2$$

for contraction blocks yields a balanced-sign threshold $\|E\|_2 < 0.1193$. The smallest chiral sign requires $\|E\|_2 < 0.00472$, and ten-percent amplitude precision there requires 4.72×10^{-4} . The weak chiral branch, rather than the balanced benchmark, sets the severe resource requirement.

Diagonal real loss merely changes the real filter and preserves the two Segre identities. Coherent diagonal phase error preserves them over $\mathbb{C}$, though a real sign still needs a phase reference. Transfer error or mismatch between emulator copies leaves the common carrier. The linear residual $\sum_T \hat{Z}_T$ is therefore the first structural null, followed by the cubic residual $\sum_T \hat{Z}_T^3$.

7 Discussion

The Golden transfer separates three levels of information. Its singular values determine every intrinsic bosonic scalar. A calibrated port frame adds permanent amplitudes and oriented control information. Antisymmetric exchange statistics retain the determinant orientation as a one-dimensional representation of the port-frame action. The Segre coordinate is therefore not an arbitrary choice of cubic observable: it is the top exterior amplitude of the transfer.

The example also separates a feasible optical certification from a direct many-particle realization. Current mesh and tomography technology can test the one-particle carrier and its bosonic shadows. Direct photonic emulation of the three-fermion amplitude depends on an additional antisymmetric resource. This boundary is part of the result rather than an experimental claim.

Verification

The exact spectra, balanced census, permanent controls, six-mode compilation, error thresholds, shot budgets, and decoding schedules are accompanied by exact generators, compact certificates, SHA-256 manifests, and independent standard-library replays in the project repository. The final archival version will identify a stable artifact release and the precise trust boundary for each displayed finite claim.

References

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